

DEXAFOS 2MG/ML INJECTION

Description and Composition

DEXAFOS 2MG/ML INJECTION is a colourless to light yellow clear solution. Each ml contains 2mg of Dexamethasone (as sodium phosphate) as its active ingredient and 15.6mg Benzyl Alcohol as its preservative.

Pharmacodynamics

Dexamethasone is a potent synthetic glucocorticoid with low mineralocorticoid activity. Dexamethasone has ten to twenty times the anti-inflammatory activity of prednisolone at an equivalent molar dose. Corticosteroids may decrease the immune response. Indeed, they inhibit capillary dilatation, leukocyte migration and phagocytosis. Glucocorticoids have an effect on metabolism by increasing gluconeogenesis. Administration of dexamethasone mimics the effects of cortisol and therefore, produces a signal that initiates the induction of parturition in ruminants if the foetus is alive.

Pharmacokinetics

After administration of the product intramuscularly, dexamethasone sodium phosphate is rapidly absorbed and hydrolysed to dexamethasone (base) giving a rapid and short-acting response (approximately 48 hours). T_{max} in cattle, horses, pigs, dogs and cats is reached within 30 minutes after intramuscular administration. $T_{1/2}$ (half-life time) varies between 5 and 20 hours depending on the species. The bioavailability after intramuscular administration is approximately 100%.

Indication

Horses, cattle, pigs, dogs and cats: Treatment of inflammation and allergic reactions.

Cattle:

Treatment of primary ketosis (acetonemia). Induction of parturition.

Recommended Dose

Horses:

Intravenous and intramuscular use.

Cattle, pigs, dogs and cats:

Intramuscular use.

For the treatment of inflammatory or allergic conditions:

The following average doses are advised. However, the actual dose used should be determined by the severity of the signs and the length of time for which they have been present.

Horses, cattle and pigs: 0.06mg of dexamethasone per kg body weight (1.5ml of product per 50kg body weight).

Dogs and cats: 0.1mg of dexamethasone per kg body weight (0.5ml of product per 10kg body weight).

For the treatment of primary ketosis:

Cattle: 0.02mg to 0.04mg of dexamethasone per kg body weight (5ml to 10ml of product per 500kg body weight) given by single intramuscular injection is advocated dependent on the size of the animal and the duration of the signs. Higher doses (i.e. 0.04mg per kg) will be required if the signs have been present for some time or if relapsed animals are being treated.

For the induction of parturition:

Cattle: To avoid foetal oversize and mammary oedema. A single intramuscular injection of 0.04mg of dexamethasone per kg body weight (10ml of product for a cow weighing 500kg) after day 260 of pregnancy. Parturition will normally occur within 48 to 72 hours.

These quantities are not specific and are quoted purely as a guide. In horses producing food intended for human consumption a total dose of 0.06mg dexamethasone per kg body weight should not be exceeded. Strict asepsis is essential.

Route of Administration

To be given by intravenous and intramuscular injection in horses.

To be given by intramuscular injection in cattle, pigs, dogs and cats.

Contraindication

Except in emergency situations, do not use in animals suffering from diabetes mellitus, renal insufficiency, cardiac insufficiency, hyperadrenocorticism or osteoporosis.

Do not use in viral infections during the viraemic stage or in cases of systemic mycotic infections.

Do not use in animals suffering from gastrointestinal or corneal ulcers, or demodicosis.

Do not use in cases of known hypersensitivity to the active substance, to corticosteroids or to any of the excipients.

See also section Pregnancy and Lactation.

Warning and Precautions

Special warnings for each target species:

None.

Special precautions for use

Special precautions for use in animals:

Response to long-term therapy should be monitored at regular intervals by a veterinary surgeon. Use of corticosteroids in horses has been reported to induce laminitis. Therefore, horses treated with such preparations should be monitored frequently during the treatment period.

Because of the pharmacological properties of the active ingredient, special care should be taken when the product is used in animals with a weakened immune system.

Except in cases of ketosis and induction of parturition, the purpose of corticosteroid administration is to induce an improvement in clinical signs rather than a cure. The underlying disease should be further investigated.

Care should be taken not to overdose Channel Island breeds of cattle.

Special precautions to be taken by the person administering the veterinary medicinal product to animals:

This product contains dexamethasone which can cause allergic reactions in some people.

People with known hypersensitivity to dexamethasone should avoid contact with the veterinary medicinal product.

Care should be taken to avoid accidental self-injection. In case of accidental self-injection, seek medical advice immediately and show the package leaflet or the label to the physician.

Dexamethasone may affect fertility or the unborn child. To avoid the risk from accidental self-injection, pregnant women should not handle this product.

This product is a skin and eye irritant. Avoid contact with skin and eyes. In the event of accidental eye or skin contact, wash/irrigate the area with clean running water. Seek medical attention if irritation persists. Wash hands after use.

Interaction with Other Medicaments

Concurrent use with non-steroidal anti-inflammatory drugs (NSAIDs) may exacerbate gastrointestinal tract ulceration.

Because corticosteroids can reduce the immune response to vaccination, dexamethasone should not be used in combination with vaccines or within two weeks after vaccination. Administration of dexamethasone may induce hypokalaemia and hence increase the risk of toxicity from cardiac glycosides. The risk of hypokalaemia may be increased if dexamethasone is administered together with potassium depleting diuretics.

Concurrent use with anticholinesterase may lead to increased muscle weakness in patients with myasthenia gravis.

Glucocorticoids antagonise the effects of insulin.

Concurrent use with phenobarbital, phenytoin and rifampicin can reduce the effects of dexamethasone.

Pregnancy and Lactation

Apart from the use of the veterinary medicinal product to induce parturition in cattle, corticosteroids are not recommended for use in pregnant animals. Administration in early pregnancy is known to have caused foetal abnormalities in laboratory animals.

Administration in late pregnancy may cause early parturition or abortion.

Use of corticosteroids in lactating cows may cause a temporary reduction in milk yield.

In suckling animals, the veterinary medicinal product should be used only according to the benefit-risk assessment by the responsible veterinarian.
See section Side Effects.

Side Effects

Anti-inflammatory corticosteroids, such as dexamethasone, are known to exert a wide range of side effects. Whilst single high doses are generally well tolerated, they may induce severe side-effects in long term use and when esters possessing a long duration of action are administered. Dosage in medium to long term use should therefore generally be kept to the minimum necessary to control symptoms.

Steroids themselves, during treatment, may cause iatrogenic hyperadrenocorticism (Cushing's disease) involving significant alteration of fat, carbohydrate, protein and mineral metabolism, e.g. redistribution of body fat, muscle weakness and wastage and osteoporosis may result.

During therapy effective doses suppress the hypothalamo-pituitary-adrenal axis. Following cessation of treatment, symptoms of adrenal insufficiency extending to adrenocortical atrophy can arise and this may render the animal unable to deal adequately with stressful situations. Consideration should therefore be given to means of minimising problems of adrenal insufficiency following the withdrawal of treatment, e.g. dosing to coincide with the time of the endogenous cortisol peak (i.e. in the morning with regard to dogs and the evening regarding cats) and a gradual reduction of dosage.

Systemically administered corticosteroids may cause polyuria, polydipsia and polyphagia, particularly during the early stages of therapy. Some corticosteroids may cause sodium and water retention and hypokalaemia upon long term use. Systemic corticosteroids have caused deposition of calcium in the skin (calcinosis cutis) and may cause atrophy of the skin.

Corticosteroids may delay wound healing and the immunosuppressant actions may weaken resistance to or exacerbate existing infections. In the presence of bacterial infection, antibacterial drug cover is usually required when steroids are used. In the presence of viral infections, steroids may worsen or hasten the progress of the disease.

Gastro-intestinal ulceration has been reported in animals treated with corticosteroids and gastrointestinal ulceration may be exacerbated by steroids in patients given non-steroidal anti-inflammatory drugs (NSAIDs) and in animals with spinal cord trauma. Steroids may cause enlargement of the liver (hepatomegaly) with increased serum hepatic enzymes.

Steroids may be associated with behavioural changes in dogs and cats (occasional depression in cats and dogs, aggressiveness in dogs). Corticosteroid use may induce changes in blood biochemical and haematological parameters. Transient hyperglycaemia can occur.

The induction of parturition with corticosteroids may be associated with reduced viability of calves, an increased incidence of retained placenta and possible subsequent metritis and/or subfertility in cattle. Corticosteroid use may increase the risk of acute pancreatitis. Other possible adverse reactions associated with corticosteroid use include laminitis and reduction in milk yield.

In very rare cases, hypersensitivity reactions may occur.

The frequency of adverse reactions is defined using the following convention:

- very common (more than 1 in 10 animals treated displaying adverse reaction(s))
- common (more than 1 but less than 10 animals in 100 animals treated)
- uncommon (more than 1 but less than 10 animals in 1,000 animals treated)
- rare (more than 1 but less than 10 animals in 10,000 animals treated)
- very rare (less than 1 animal in 10,000 animals treated, including isolated reports).

Symptoms and Treatment of Overdose

An overdose can induce drowsiness and lethargy in horses. See section Side Effects.

Storage Condition

Store below 30°C.

Shelf life

3 years.

Following withdrawal of first dose, use the product within 24 hours. Any unused material should be discarded.

Withdrawal Periods

Cattle:

Meat and offal: 8 days

Milk: 72 hours

Pigs:

Meat and offal: 2 days

Horses:

Meat and offal: 8 days

Not authorised for use in horses producing milk for human consumption.

Packing

20ml, 50ml, 100ml, 250ml.

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